E35: solver B's pivot schedule, learned per curve and replayed with a counted check

Run report and results, 2026-11-02. Research note `research/notes/index-calculus/RESEARCH_GLV_INVARIANT_FACTOR_BASES.md`: §9.40 (pre-registration, commit `038e2562d3`, before any code) and §9.42 (results). Code `b1f5df9a45` (module `src/cryptanalysis/chain_trace.rs`, examples `e35_stage_a`, `e35_checks`, `e35_tables`) and `2308fa4664` (dispatch hook in `klein_v4.rs`, `quartic_folds.rs`, `mod.rs`). Binaries were built from `git archive 2308fa4664` and pinned by sha256 in `experiments/39_chain_trace/binary_sha256.txt`; the checks example was rebuilt once (sha256 `ed21fbd0…`). Records are in `experiments/39_chain_trace/`: `stage_a.log` (`stage_a.json` sha256 `6c7c8a44…`; the artifact-store upload is blocked without the AWS CLI), `checks*.{json,log}`, `identity_existing_solvers.json`, `e2e_chain-sparsest-trace_karatsuba_p*s*.json` and the log, `tables.md` and `summary.json`. Host: the 4-vCPU Claude Code container, at most two cores. isolab-eks is not reachable from it (no AWS CLI, no kubectl). No isolation tier is earned: operation counts only, no wall-time claim.

Question and design

Solver B (sparsest pivots) multiplies 1.82–1.86 × less than solver A, E31′s ranked chain solve, at 0 GAE. Its counted selection costs $1.05 \cdot 10^6$ native operations a call (E33), so it is unranked. If B's choices were a property of the curve, they could be learned once and replayed. This is the idea of Joux–Vitse's F4 trace reuse (lead atlas `KR-IC-f3da82`). On the binary S_3 line the trace-reuse engines gave 1.4–2.3 × and that lead is closed dominated (`LEAD-ICE-boolean-f4-f5-f6-engines`). E21 (§9.11) replayed a shape trace on this family's `D4` solve and lost.

T (`chain-sparsest-trace`). The trace is learned from three targets of a separate stream (B′'s elimination, B's choices, per-column mode), out of sample. In every contested column the replay does one lookup and two tests: the traced row must sit at its traced position among the sorted candidates, and its entry must be nonzero. On a miss it takes A's choice at no extra cost. A different matrix shape replays nothing. Learning (multiplications, entries, B′ selection, mode) is charged inside the record's totals.

What ran

- **Stage 0 + Stage A:** E31/E33′s calls exactly (E25′s ten GLS curves, $p \in \{101, 149, 197, 269, 397\}$, seeds 1 and 2, the first 1,000 targets of arm A's stream), one thread. Five solvers per call: E25, A counted, B counted, B″ and T, 50,000 solves. Stage 0 records B's pivot sequence on every call and compares it with the curve's trace and with the other curves' traces.
- **Controls:** 32 calls each of $p = 101$ seed 2 and $p = 397$ seed 1. Three corruptions (a used row, another pivot row, another shape), each with and without the check.
- **Identity:** E33′s pinned binary against E35′s, for seven existing configurations at $p = 101$ seed 2, and against the stored E27/E33 records.
- **End to end:** E33′s command with `--solver chain-sparsest-trace --karatsuba`, ten curves, both arms (20 rows), 0 failures.

Measured

- **Stage 0.** No call's pivot sequence equals its curve's trace, and 997–1,000 of 1,000 sequences per curve are distinct. In contested columns (334 a call), B's row equals the trace's in 69.8–88.4%, rising with p. Another curve's trace agrees as well, often better (69.3–90.2%). The stable part is generic structure; the rest depends on the target.
- **Stage A.** T agrees with E25 on all 10,000 calls. It does 744–817 selection operations a call and misses 55.6–77.5% of contested columns; misses cascade. C median: T $5.23–5.51 \cdot 10^5$, B $3.04–3.33 \cdot 10^5$, A $6.01–6.34 \cdot 10^5$. T keeps 18.8–36.8% of B's saving over A. Per-call S_A/S_T is 1.092–1.254 / 1.014–1.049 / 0.9916–0.9952 at 0 / 0.1 / 1 GAE.
- **Controls.** Every corruption was rejected (63/63 per kind) with unchanged outputs. Without the check every corrupted call aborted. A first version counted one no-op corruption as undetected (kept as `checks_v1`).
- **Identity.** Old = new for all seven configurations, and = stored for the four Karatsuba ones. All 20 T records equal E33′s B-counted records outside the registered fields: every relation, log and walk is the same.
- **End to end** (ranges over 20 rows of S_X/S_T at 0 / 0.1 / 1 GAE): A 1.093–1.252 / 1.020–1.055 / 0.9981–1.0028 (6 rows below 1 at 1 GAE); E25 2.73–3.20 / 1.421–1.507 / 1.0325–1.0557; B″ 0.709–0.807 / 0.975–1.007 / 1.053–1.059; B 0.60–0.68 / 5.19–5.48 / 6.55–6.74. Distance to the matched rho (walk (iii) of E27), S_T/rho: $5.11 \cdot 10^3$–$3.53 \cdot 10^5$ / $2.02 \cdot 10^4$–$1.35 \cdot 10^6$ / $9.04 \cdot 10^4$–$6.63 \cdot 10^6$. Rho is ahead of every row at every weight.

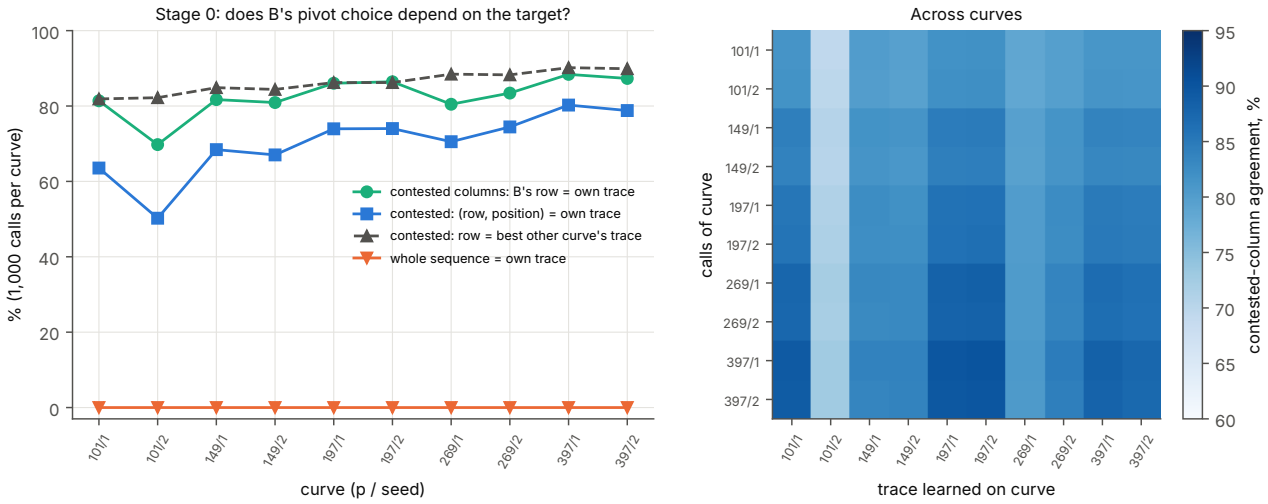


Figure 1: Stage 0. Left: per curve over 1,000 calls, the share of contested columns where B's pivot row (and position) equals the curve's trace, the same against the best other curve's trace, and whole-sequence equality (0 everywhere). Right: contested-column agreement of each curve's calls with each curve's trace. Source: `summary.json`.

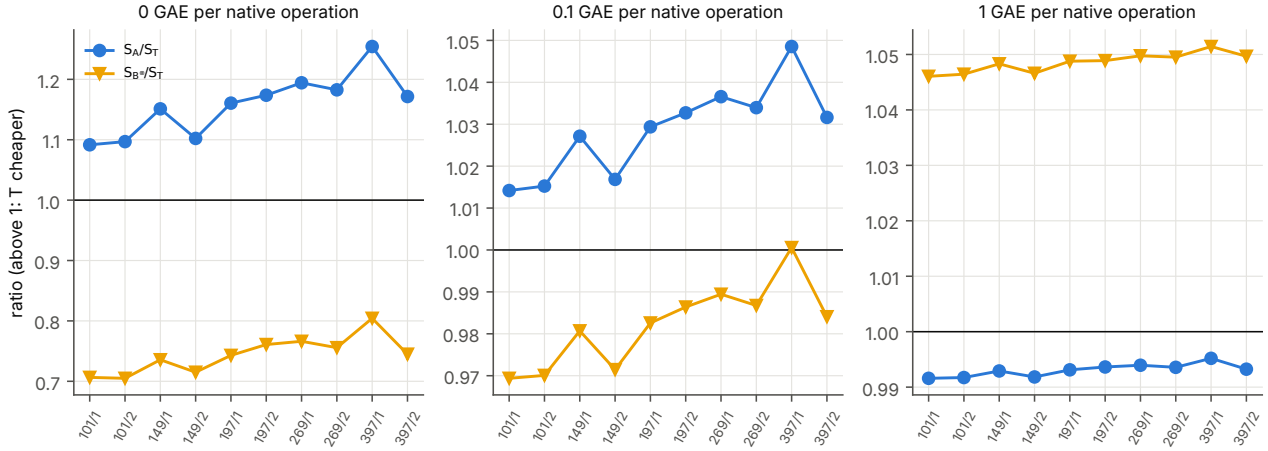


Figure 2: Stage A per-call mean $S(g) = C + 97g(\text{entries} + \text{selection})$ of A and B″ over T's (learning amortised over the 1,000 calls), per curve. Above 1, T is cheaper. T is behind A at 1 GAE on every curve and behind B″ at 0 and 0.1 GAE. Source: `summary.json`.

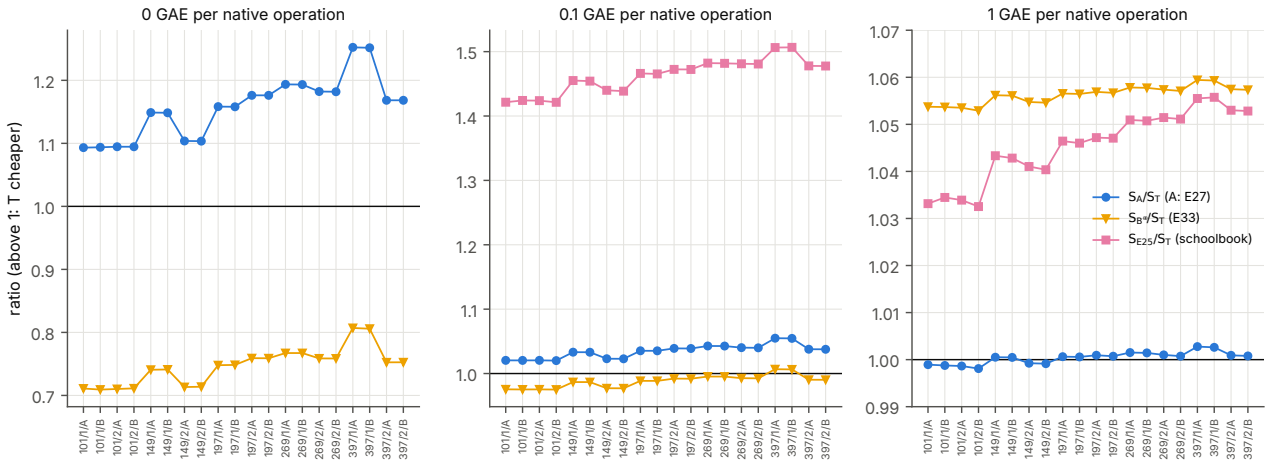


Figure 3: End to end, per row (p/seed/arm): S_X/S_T for A (E27′s records), B″ (E33′s) and E25 (schoolbook; omitted at 0 GAE, 2.7–3.2). B and B′ are far off scale (0.6–0.68 at 0 GAE; 2.1–6.7 above). At 1 GAE, T is ahead of A on 14 rows and behind on 6. Source: `summary.json`.

Verdicts (pre-registered, §9.40)

G2.145, G2.146 (controls v2) and G2.147 hold. G2.144 is **falsified**: contested-column agreement is below 90% on every curve. G2.148, G2.149, G2.150, G2.151 (bands), G2.152 and G2.153 (upper end of the range) are **falsified**.

Ranks (only where the order holds at 0, 0.1 and 1 GAE on every row)

- T against A: **unresolved**. T is ahead at 0 and 0.1 GAE on every row; at 1 GAE it is ahead on 14 rows and behind on 6. A remains the ranked solver.
- T against E25: ahead at every weight on every row, as A already is (§9.24). This is not new.
- T against B, B′ and B″: **unresolved**. Each is ahead of T at 0 GAE; B″ is also ahead at 0.1 GAE on 18 of 20 rows; T is ahead of all three at 1 GAE.
- Rho is ahead of T across the band. Distance remaining at 0 GAE, best row: $5.11 \cdot 10^3\ \times$ (T, unranked); A, the ranked solver, is at $6.39 \cdot 10^3\ \times$.

What it says, and what it does not

There is no per-curve pivot order to replay. B's tie-breaks among rows with nearly equal counts are set by accidental zeros, so they change with the target. A counted replay keeps B's choice in only 23–44% of contested columns. It is break-even with A at 1 GAE and behind B″ at 0 and 0.1 GAE, so it is **not an improvement** under the 2026-10-15 criterion. Class: engineering, negative. It agrees with E21 on a different object. Not tested: a membership test through a row-to-lead map instead of the position test. It could recover at most the $\approx 2/3$ of misses whose row was still a candidate, and its bookkeeping would have to be priced (exploratory, not registered). The next lever is unchanged: the Macaulay matrix itself.

Limits

- Operation counts on a shared container; no wall time, no isolation tier.
- In-crate unit tests were not added or run: a release test build needs about 9 GB. The solver-level checks run as examples (`e35_checks`, Stage A identity), and the library and tests type-check (`cargo check`).
- S_{E25}/S_T prices T's learning in Karatsuba arithmetic (about $5 \cdot 10^5$ multiplications per run, $< 10^{-4}$ of S).
- The positive control aborts partly because of the trace's own invalid entries, not the corruption alone.
- Not validated: no replay certificate or validator report, so no leaderboard or `benchmarks/` row.